1. Introduction: Why Comparison Matters (0–10 minutes)

- Rapid growth of immersive technologies.
- Incorrect technology choice leads to high cost and low usability.
- Need to understand **where AR fits vs VR vs MR**.

📌 **Example:**
Using VR for on-site machine maintenance vs AR for overlaying repair instructions.

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2. AR vs VR vs MR – Conceptual Overview (10–20 minutes)

Definitions Recap:

- **AR:** Enhances real world with digital content.
- **VR:** Replaces real world with virtual environment.
- **MR:** Blends real and virtual worlds with interaction.

Core Difference:

- Level of **immersion**
- Level of **real-world awareness**

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3. Detailed Comparison: AR vs VR vs MR (20–35 minutes)

Feature	AR	VR	MR
Real World Visibility	Yes	No	Yes
Immersion Level	Low–Medium	High	Medium–High
User Isolation	No	Yes	Partial
Interaction	Limited	Full	Advanced
Hardware	Mobile, Glasses	Headsets	HoloLens
Cost	Low	Medium–High	High
Use Environment	Real world	Virtual world	Hybrid

📌 **Examples:**

- AR → Google Maps Live View
- VR → Flight Simulator
- MR → HoloLens industrial training

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4. Selection of Technology: AR or VR (35–50 minutes)

Key Decision Factors:

1. Application Objective

- Visualization → AR
- Simulation → VR

2. User Environment

- Real-world operation → AR
- Controlled environment → VR

3. Level of Immersion Required

- Low → AR
- High → VR

4. Hardware Availability

- Mobile devices → AR
- Dedicated headsets → VR

5. Cost and Scalability

- Low budget & scalability → AR
- High-end experience → VR

📌 **Case Example:**
Medical training → VR
On-site equipment repair → AR

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5. Case Discussion & Examples (50–55 minutes)

Case 1: Education

- AR for classroom visualization
- VR for virtual laboratories

Case 2: Industry

- AR for assembly guidance

- VR for safety training

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6. Recap & Student Interaction (55–60 minutes)

Key Takeaways:

- AR adds, VR replaces, MR blends
- Selection depends on use case, cost, and immersion
- No single technology fits all problems

Important Questions (Exam-Oriented)

1. **Compare Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR) with suitable examples.**
   (8–10 marks)
2. **Explain the factors involved in selecting AR or VR technology for an application.**
   (8 marks)
3. **Differentiate between AR and VR in terms of immersion, interaction, and hardware.**
   (5 marks)